Lecture 5, Mar 25 2026.

Problem For what values of r is the sequence $\{r^n\}$ convergent?

Sln. for $r=1$ $\{r^n\}_{n=1}^{\infty} = \{1,1,1,1,\cdots\}$ which is convergent.

$\lim_{x\to\infty} b^x = \infty$ for $b>1$ and $\lim_{x\to\infty} b^x = 0$ for $0<b<1$.

let $b=r$ then

$$\lim_{n\to\infty} r^n = \begin{cases} \infty & \text{if } r>1. \\ 0 & \text{if } 0<r<1. \end{cases}$$

$\lim_{n\to\infty} |r^n| = \lim_{n\to\infty} |r|^n = 0$ if $-1<r<0$ Then, by the thm.

$$\lim_{n\to\infty} r^n = 0$$

So $\{r^n\}$ is convergent for $-1<r\leq 1$.

Def A sequence $\{a_n\}$ is called increasing if $a_n < a_{n+1}$ for all $n\geq 1$. that is $a_1 < a_2 < a_3 < \cdots$. It is called decreasing if $a_n > a_{n+1}$ for all $n\geq 1$. A sequence is called monotonic if it is either decreasing or increasing.

Example. The sequence $\left\{\frac{3}{n+5}\right\}$ is decreasing.

$$a_n = \frac{3}{n+5} \qquad a_{n+1} = \frac{3}{n+1+5} = \frac{3}{n+6}$$

Since $n+6 > n+5$ then $\frac{3}{n+5} > \frac{3}{n+6}$ i.e., $a_n > a_{n+1}$.

So. a_n is decreasing.

Other solution. Let $f(x) = \frac{3}{x+5}$ and $f'(x) = -\frac{3}{(x+5)^2} < 0$ So f is decreasing for all $x\geq 1$.

Example. Show that $a_n = \frac{n}{n^2+1}$ is decreasing.

Sln. Note that

$$a_n = \frac{n}{n^2+1} \quad \text{and} \quad a_{n+1} = \frac{n+1}{(n+1)^2+1} = \frac{n+1}{n^2+2n+2}$$

WNS: $\frac{n}{n^2+1} > \frac{n+1}{n^2+2n+2}$

$$\Leftrightarrow \quad n(n^2+2n+2) > (n+1)(n^2+1)$$

$$\Leftrightarrow \quad \cancel{n^3}+2n^2+2n > \cancel{n^3}+n+n^2+1.$$

$$\Leftrightarrow \quad n^2+n > 1.$$

Since $n \geq 1$ then $n^2+n > 1$ is true so, $a_n > a_{n+1}$.

Def. A sequence $\{a_n\}$ is bounded above if there is a number M s.t $a_n \leq M$ for all $n \geq 1$.

For example $a_n = \frac{1}{n}$ and $M = 1$ then $\frac{1}{n} \leq 1$.

A sequence $\{b_n\}$ is bounded below if there is a number m s.t. $m \leq b_n$ for all $n \geq 1$.

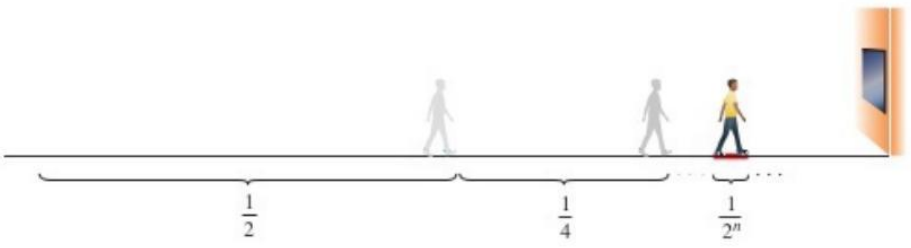
For example $b_n = \frac{1}{n}$ and $m = 0$ then $0 \leq \frac{1}{n}$.

If a sequence is bounded above and below, then it's called a bounded sequence.

Thm. (Monotonic sequence Thm). Every bounded, monotonic sequence is convergent.

11.2. Series. Let consider the example.

At the nth stage the man walks a distance $1/2^n$.



of course, we guess that the total distance is 1.

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots + \frac{1}{2^n} + \cdots = 1.$$

What means "infinite sum"?

Infinite Series. If we try to add the terms of an infinite sequence $\{a_n\}$ we get an expression

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

which is called an infinite series (or just a series) and is denoted by $\sum_{n=1}^{\infty} a_n$ or $\sum a_n$.

In general, does it make sense to talk about the sum of infinitely many numbers?

$$1 + 2 + 3 + 4 + 5 + \cdots + n + \cdots$$

this is not defined!

However, consider the series of distances from Zeno's paradox.

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \cdots + \frac{1}{2^n} + \cdots$$

then let define

$$S_n = \sum_{k=1}^{n} \frac{1}{2^k} \qquad \text{we guess that} \quad \lim_{n\to\infty} S_n = 1.$$

Notice that $\lim_{n\to\infty} S_n = \sum_{k=1}^{\infty} \frac{1}{2^k} = 1$.

Let consider $\{a_n\}_{n=1}^{\infty}$ and $S_n = \sum_{k=1}^{n}$ and.

$$S_1 = a_1.$$

$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$$\vdots$$

$$S_n = a_1 + a_2 + a_3 + \cdots + a_n = \sum_{k=1}^{n} a_n.$$

Def. the series $\sum_{n=1}^{\infty} a_n$ is convergent if $\{S_n\}_{n=1}^{\infty}$ the sequence of the partial sums is a convergent sequence if $\{S_n\}_{n=1}^{\infty}$ is divergent,

then $\sum_{n=1}^{\infty} a_n$ is divergent series.

Examples i). Let consider a series $\sum_{n=1}^{\infty} a_n$ s.t.

$$S_n = a_1 + a_2 + \cdots + a_n = \frac{2n}{3n+5}$$

So, $\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \left(\frac{2n}{3n+5}\right) = \frac{2}{3}$.

i). Show that the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ is convergent, and find its sum.

Sln.

$$S_n = \sum_{k=1}^{n} \frac{1}{k(k+1)} = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \cdots + \frac{1}{n(n+1)}$$

We can simplify using partial fractions.

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

$$S_n = \sum_{k=1}^{n} \frac{1}{k(k+1)} = \sum_{k=1}^{n} \left\{\frac{1}{k} - \frac{1}{k+1}\right\}$$

$$= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right)$$

$$= 1 - \frac{1}{n+1}.$$

So $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right) = 1.$

This series are called telescoping series.

Sum of a Geometric Series.

An important series is the geometric series.

$$a + ar + ar^2 + ar^3 + ar^4 + \cdots + ar^n + \cdots = \sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=0}^{\infty} ar^n$$

Notice that $S_n = a + ar + \cdots + ar^{n-1}$

$$rS_n = ra + ar^2 + \cdots + ar^n$$

$$S_n - rS_n = a - ar^n$$

$$S_n(1-r) = a(1-r^n)$$

$$S_n = \frac{a(1-r^n)}{1-r}$$

If $-1 < r < 1$ we know that $r^n \to 0$ as $n \to \infty$ so.

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r} = \frac{a}{1-r} \quad \text{for } |r| < 1.$$

and is divergent for $|r| \geq 1$.

The geometric series $\sum_{n=0}^{\infty} ar^n = \sum_{n=1}^{\infty} ar^{n-1}$ is convergent for $|r| < 1$.

and its sum is $\sum_{n=0}^{\infty} ar^n = \sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$

Example. Find the sum of the geometric series.

$$5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \cdots$$

Sln. $\sum_{n=0}^{\infty} 5(-1)^n \left(\frac{2}{3}\right)^n = \sum_{n=0}^{\infty} 5\left(-\frac{2}{3}\right)^n$ is geometric series with $a=5$,

$r = -\frac{2}{3}$ and $|r| = \frac{2}{3} < 1$ so the series is convergent and.

$$\sum_{n=0}^{\infty} 5\left(-\frac{2}{3}\right)^n = \frac{5}{1-\left(-\frac{2}{3}\right)} = 3.$$

Example. Is the series $\sum_{n=1}^{\infty} 2^{2n} 3^{1-n}$ convergent or divergent?

Sln.

$$\sum_{n=1}^{\infty} 2^{2n} 3^{1-n} = \sum_{n=1}^{\infty} (2^2)^n \cdot \left(\frac{1}{3}\right)^{n-1} = \sum_{n=1}^{\infty} \frac{4^n}{3^{n-1}} = \sum_{n=1}^{\infty} 4 \cdot \frac{4^{n-1}}{3^{n-1}}$$

$$= \sum_{n=1}^{\infty} 4\left(\frac{4}{3}\right)^{n-1}$$ this is a geometric series with

$$a = 4 \quad r = \frac{4}{3}.$$

and $|r| = \frac{4}{3} > 1$ then the series is divergent.